

11-matching-flow: 匹配与网络流 (Matching and Network Flow)

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3 Theorems + 1 Algorithm

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



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Theorem

$\mathcal{S}$ is maximum *iff* G does not contain $\mathcal{S}$ -augmenting objects.

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Algorithm

Repeatedly finding $\mathcal{S}$ -augmenting objects until no more ones exist.

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



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To **minimize** the size of its **dual** mathematical structure $\mathcal{S}'$ in G

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Theorem (Weak Duality Theorem)

The size of a maximum $\mathcal{S} \leq$ The size of a minimum $\mathcal{S}'$

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



To **minimize** the size of its **dual** mathematical structure $\mathcal{S}'$ in G

Theorem (Weak Duality Theorem)

The size of a maximum $\mathcal{S} \leq$ The size of a minimum $\mathcal{S}'$

Theorem (Strong Duality Theorem)

The size of a maximum $\mathcal{S} =$ The size of a minimum $\mathcal{S}'$

let's get
married
today

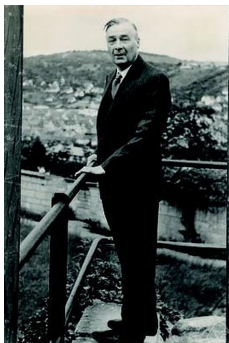


The Marriage Problem (Philip Hall, 1935)

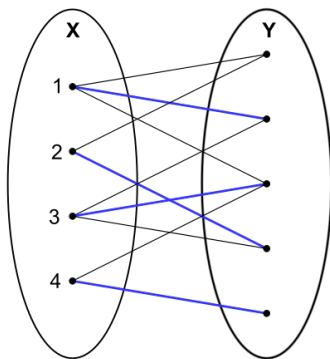
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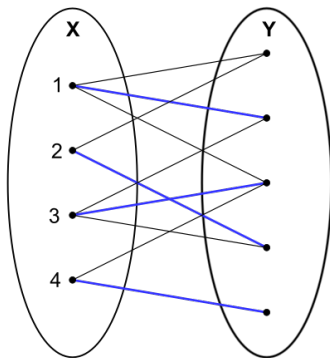
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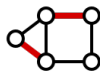
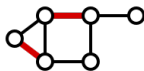
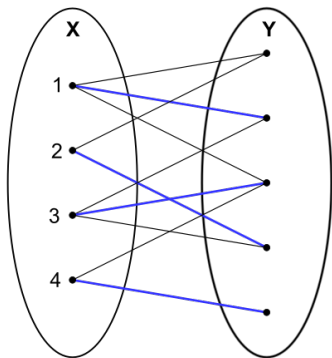
Philip Hall (1904 ~ 1982)





Definition (Matching (匹配))

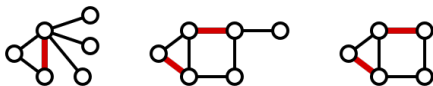
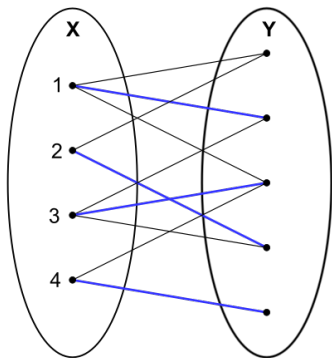
A **matching** in a graph G is a set of edges with **no shared endpoints**.



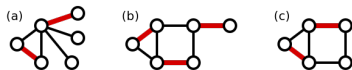
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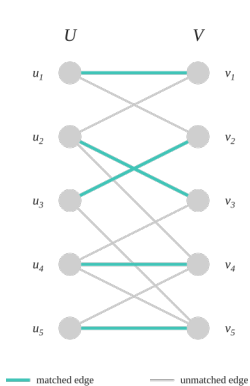
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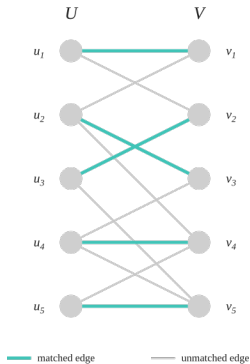
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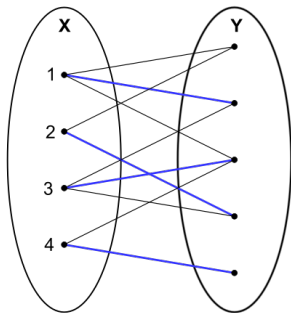
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Perfect Matching



Perfect Matching



X -Perfect Matching

Definition (X -Perfect Matching (X -Saturating Matching))

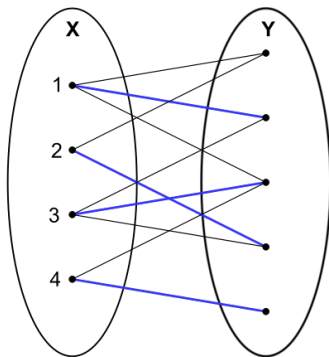
Let $G = (X, Y, E)$ be a bipartite graph.

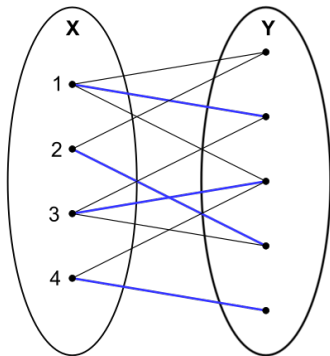
An X -perfect matching of G is a matching which covers each vertex in X .

The Marriage Problem (Philip Hall, 1935)

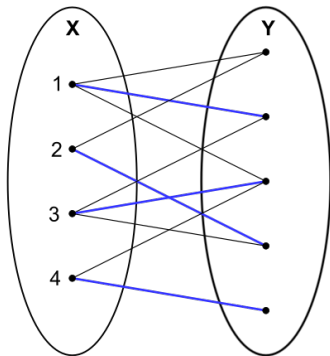
Let $G = (X, Y, E)$ be a bipartite graph.

Is there an **X-perfect matching** of G ?

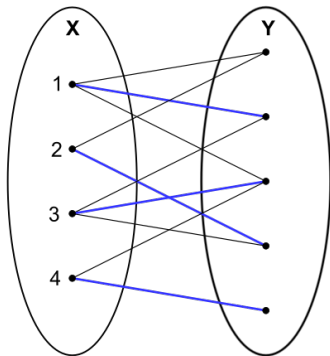




$$|X| \leq |Y|$$



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$$\forall W \subseteq X. |W| \leq |N(W)|$$

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

There is an *X -perfect matching* of G if and only if

$$\forall W \subseteq X \quad |W| \leq |N_G(W)|.$$

By induction on the number $|X|$ of vertices in X .

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How to reduce the case $|X| = m$ to a case $|X'| < m$?

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Therefore, there is an X -perfect matching in G .

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Definition (M -augmenting Paths)

An M -augmenting path is an M -alternating path whose endpoints are unsaturated by M .

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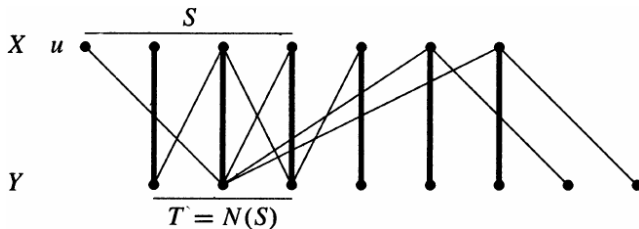
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Let M be a *maximum* matching in G .

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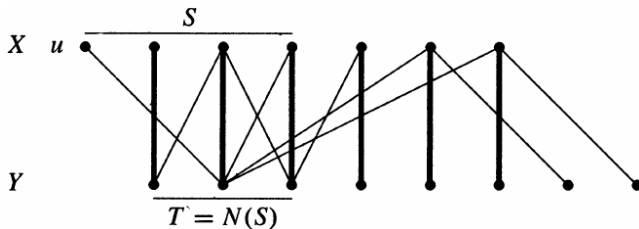
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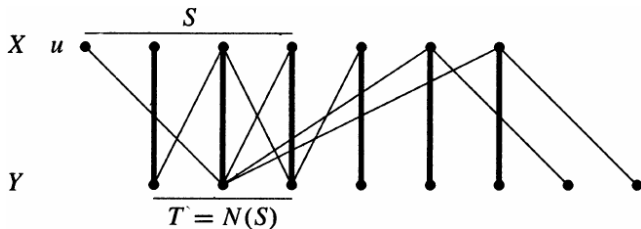


We show that Hall's Condition is violated for *some* $S \subseteq X$.

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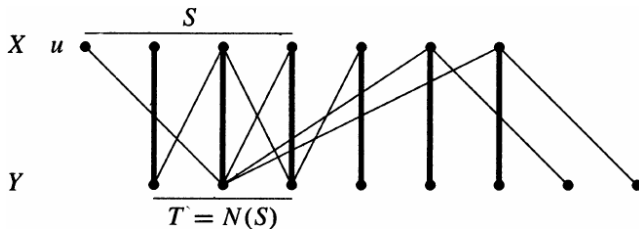
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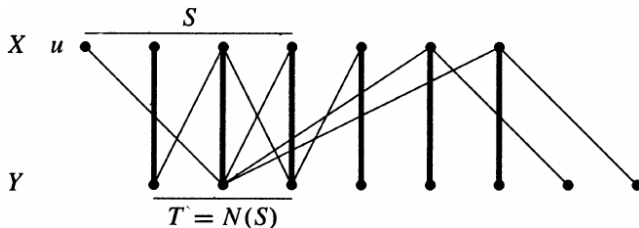
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We will construct $S \subseteq X$ based on M .

Let $u \in X$ be a vertex of X *not* saturated by M .

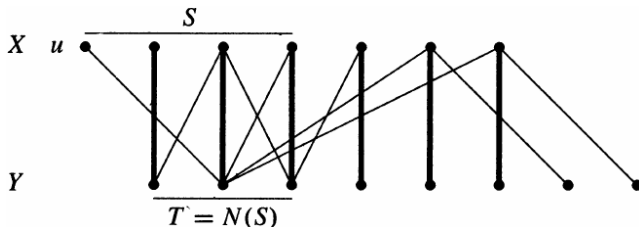


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Consider all *M -alternating paths* starting from u .

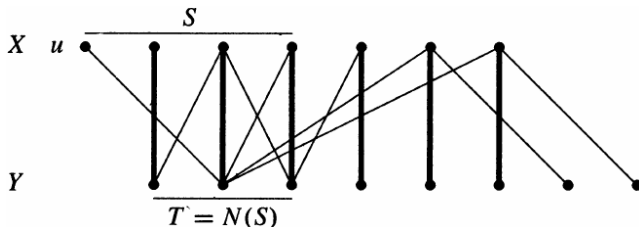
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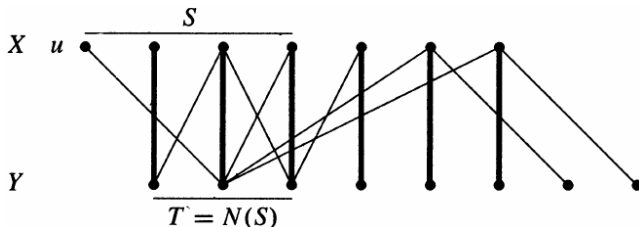


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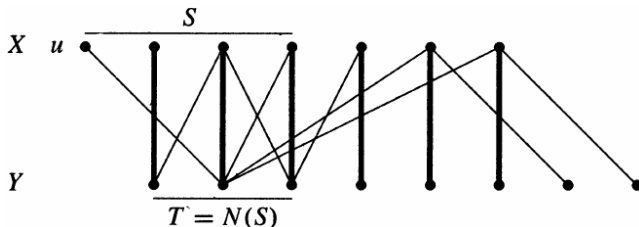
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We will show that

$$T = N(S) \wedge |T| = |S - \{u\}|$$

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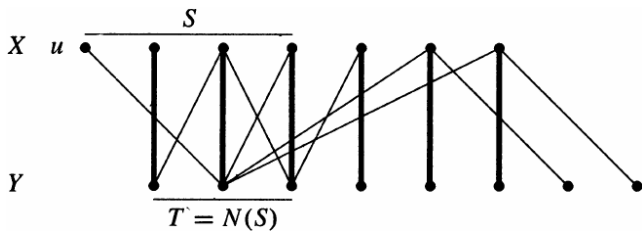
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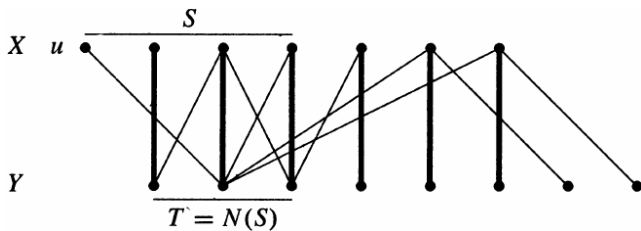
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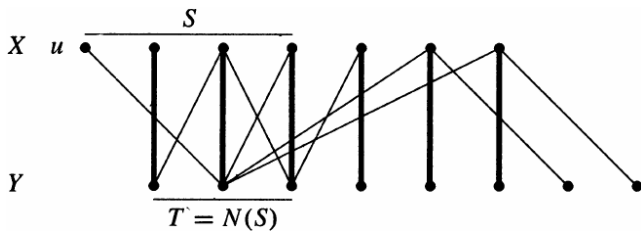


$$|T| = |S - \{u\}|$$



We show that there is a **bijection** from T to $S - \{u\}$.

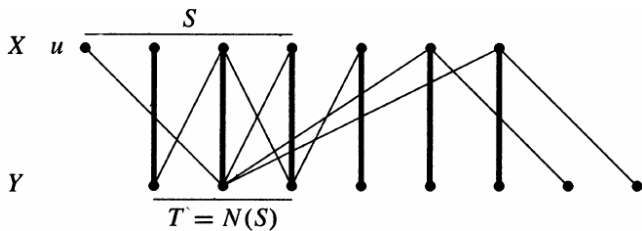
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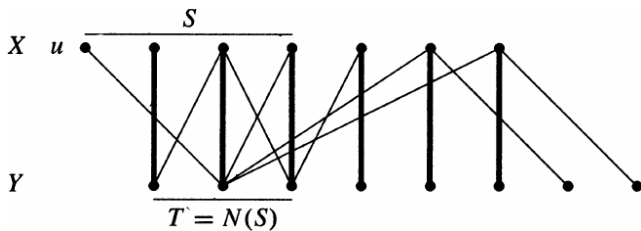


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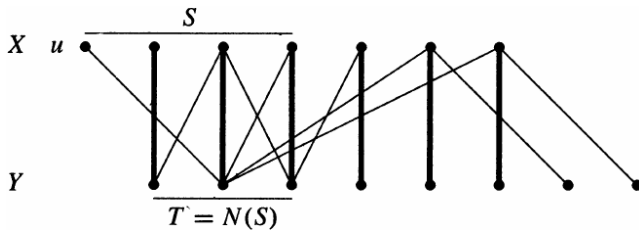
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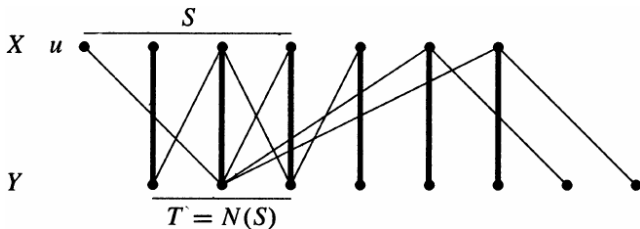
Every vertex in T goes via an edge in **M** to a vertex in $S - \{u\}$.

Every vertex in $S - \{u\}$ is reached by an edge in **M** from a vertex in T .

$$T = N(S)$$

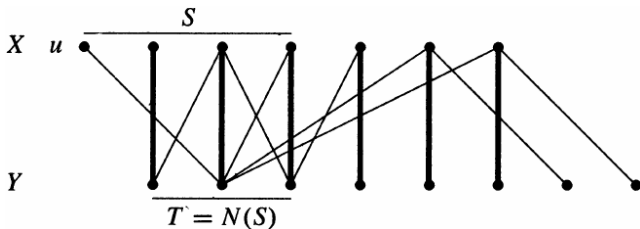


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$$M \text{ matches } T \text{ with } S - \{u\} \implies T \subseteq N(S)$$

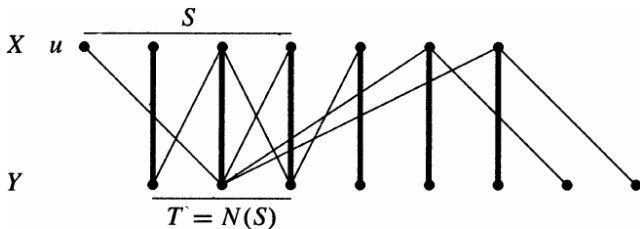
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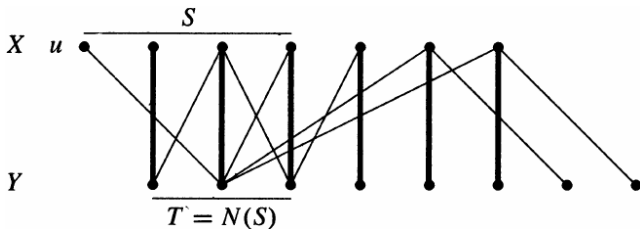


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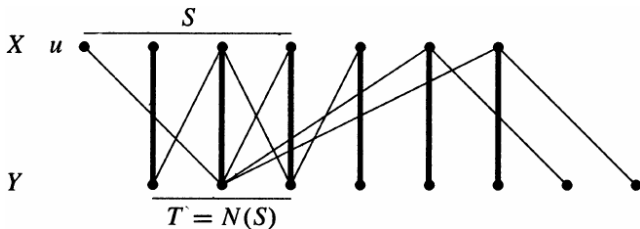
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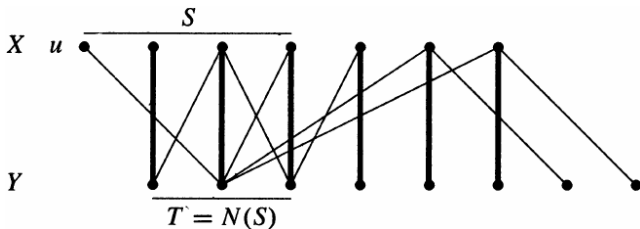
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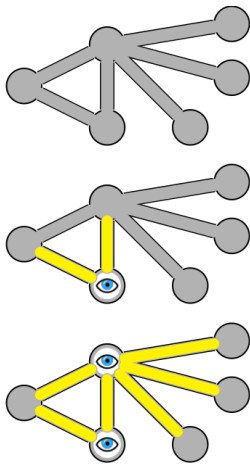
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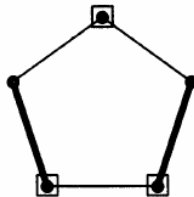
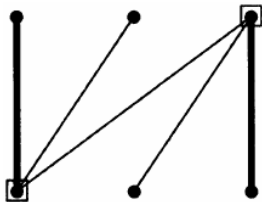
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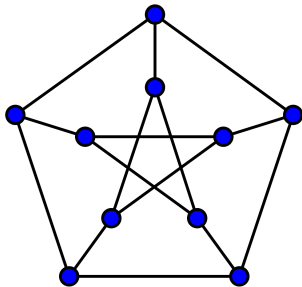
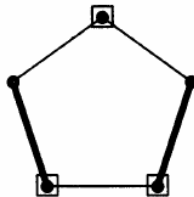
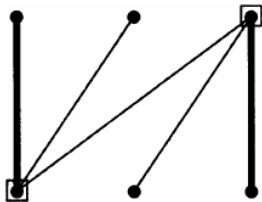
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Definitions (Vertex Cover (点覆盖))

A **vertex cover** of a graph G is a set $Q \subseteq V(G)$ that **covers** all edges.





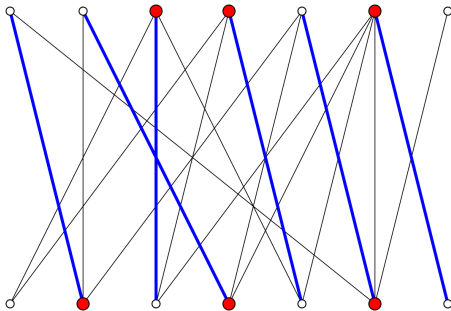


Theorem (Weak Duality Theorem (弱对偶定理))

*Let G be a graph. The maximum size of a matching in G
 $\leq$ the minimum size of a vertex cover of G .*

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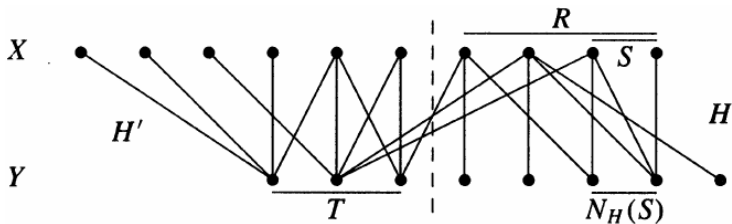


Theorem (König (1931), Egerváry (1931))

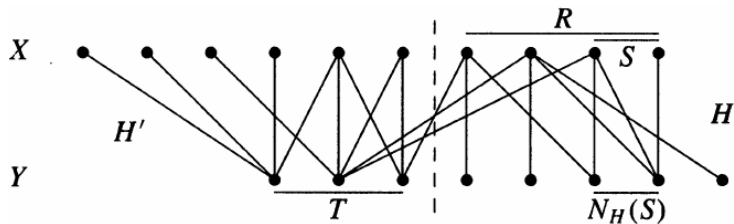
Let G be a *bipartite* graph. The maximum size of a matching in G
 $=$ the minimum size of a vertex cover of G

Given a minimum vertex cover Q ,
we construct a matching M of the same size.

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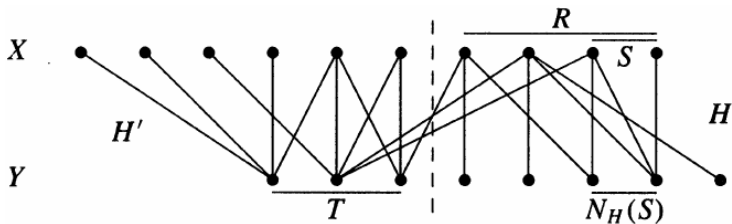


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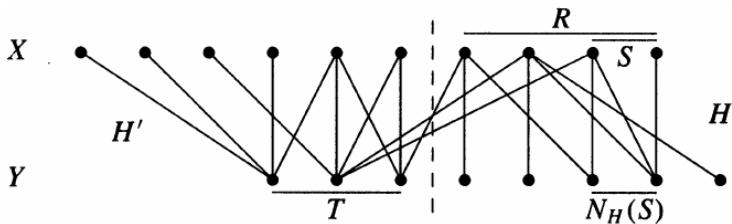


$$R = Q \cap X \quad T = Q \cap Y$$

$H \triangleq (R \cup (Y - T))$ -induced subgraph of G

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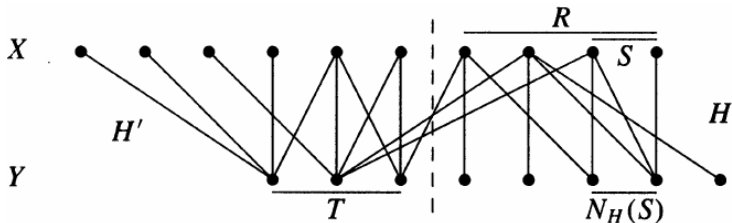
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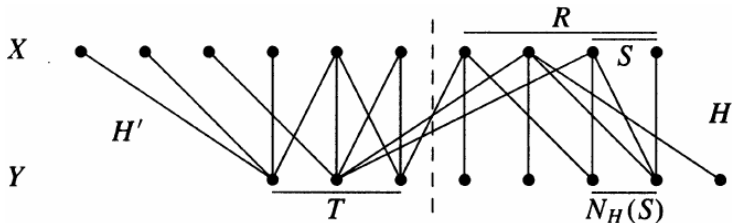
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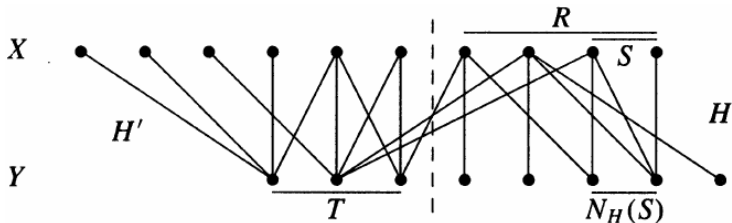


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To show that Hall's Condition holds for H (and R).

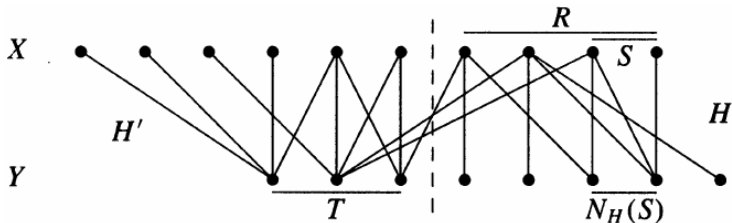
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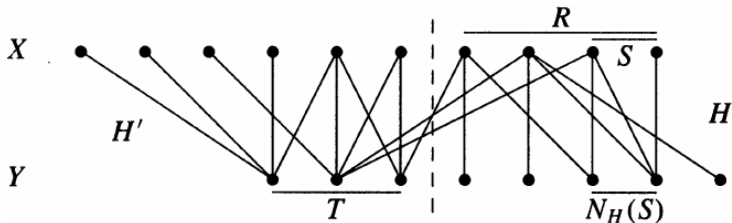


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$$\exists S \subseteq R. |N_H(S)| < |S|$$

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$T \cup (R - S + N_H(S))$ is a smaller vertex cover than Q .

Augmenting Path Algorithm for Maximum Bipartite Matching

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► $M = \emptyset$

Augmenting Path Algorithm for Maximum Bipartite Matching

- ▶ $M = \emptyset$
- ▶ **While** there **is** an M -augmenting path P :
 - ▶ Enlarge M by P : $M \leftarrow M \oplus P$

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How to **find M -augmenting paths** efficiently?

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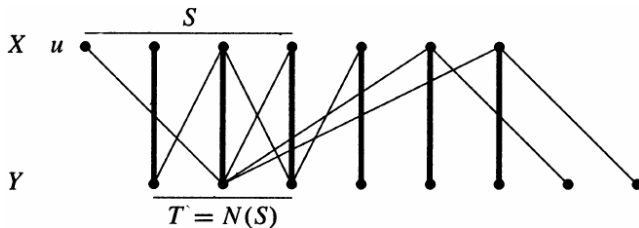
How to prove that **the M returned** is a **maximum** matching?

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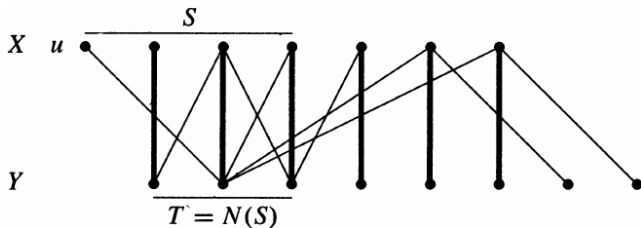
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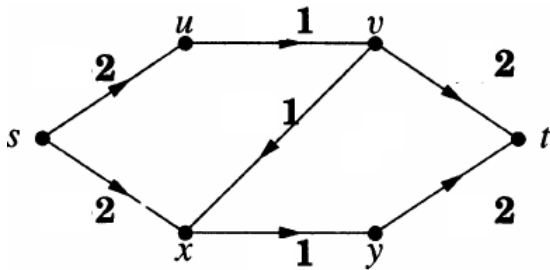


When the algorithm terminates,
it actually finds a **vertex cover** $T \cup (X - S)$ of size M .

Definition (Network (网络))

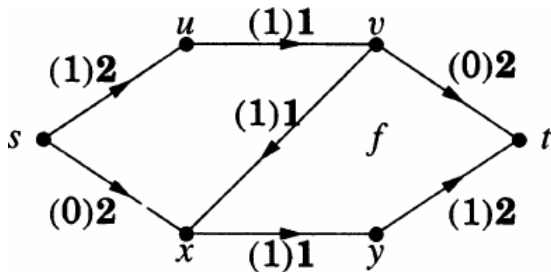
A **network** is a **digraph** with

- ▶ a distinguished **source vertex** s ,
- ▶ a distinguished **sink vertex** t ,
- ▶ a **capacity** $c(e) \geq 0$ on each edge e



Definition (Flow (流))

A **flow** f is a **function** that assigns a value $f(e)$ to each edge e .



Definition (Feasible Flow (可行流))

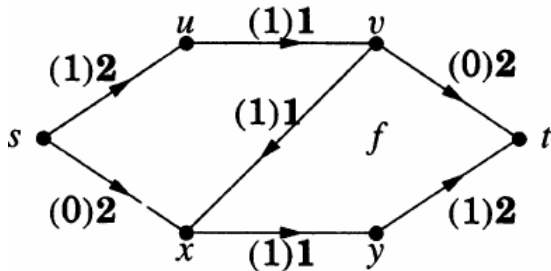
A flow f is **feasible** if it satisfies

Capacity Constraints:

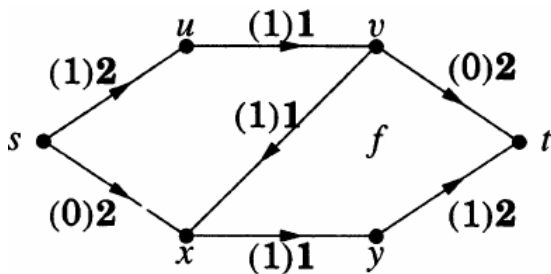
$$\forall e \in E. 0 \leq f(e) \leq c(e)$$

Flow Conservation:

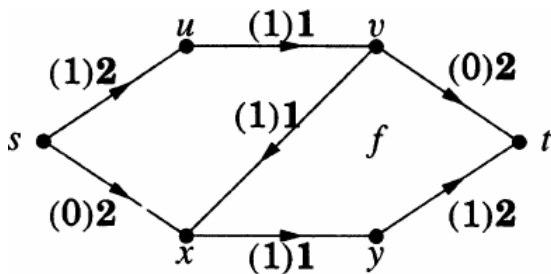
$$\forall v \in V. f^+(v) = f^-(v)$$



$$f^+(v) = \sum_{(v,w) \in E} f(v,w) \quad f^-(v) = \sum_{(u,v) \in E} f(u,v)$$

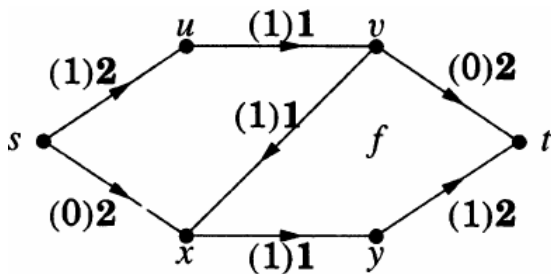


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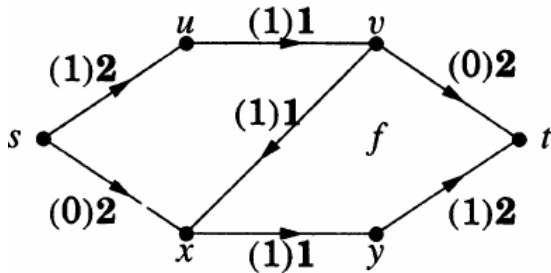


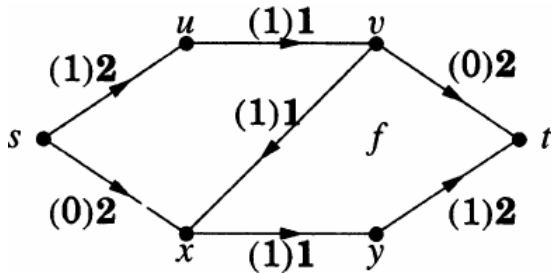
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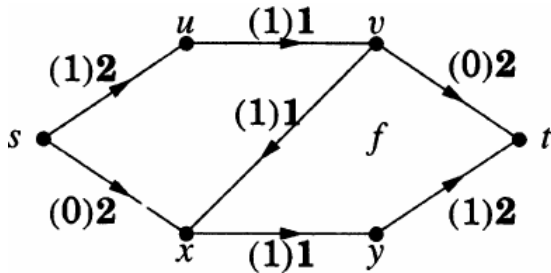


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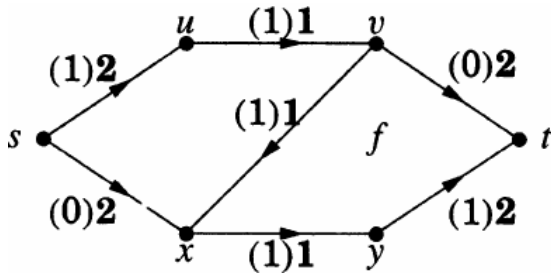


$$\forall U \subseteq (V - \{s, t\}). f^+(U) = f^-(U)$$



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$$s \in U \wedge t \notin U \implies f^+(U) - f^-(U) =$$



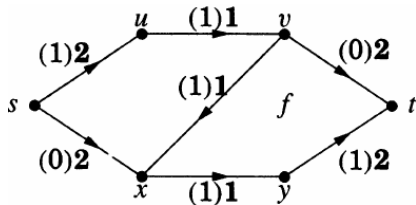
$$\forall U \subseteq (V - \{s, t\}). f^+(U) = f^-(U)$$

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Definition (Value (值))

The **value** $\text{val}(f)$ of a **flow** f is

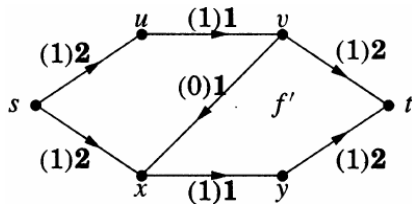
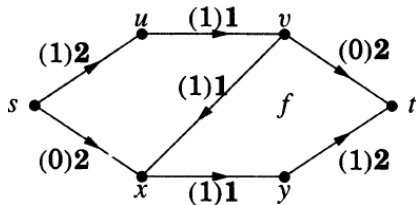
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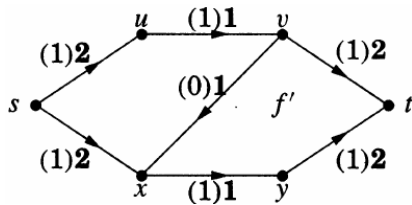
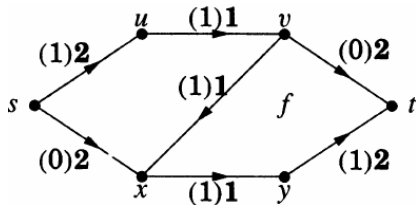
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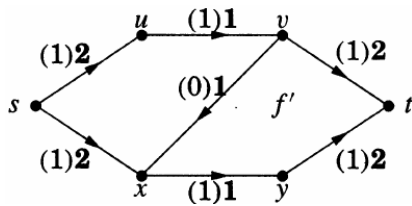
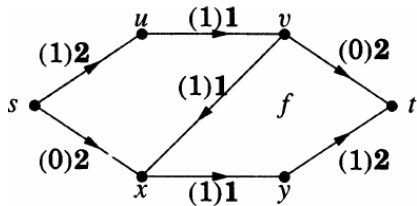
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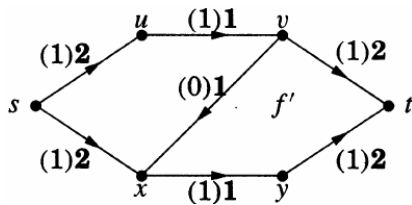
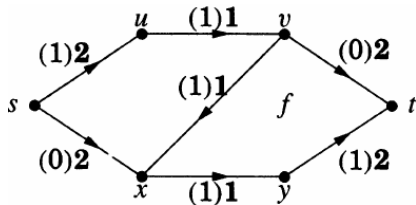
Definition (Maximum Flow (最大流))

A **maximum flow** is a **feasible flow** of maximum **value**.

$$s - x - v - t$$



$$s - x - v - t$$



Definition (f -augmenting Paths (增广路径))

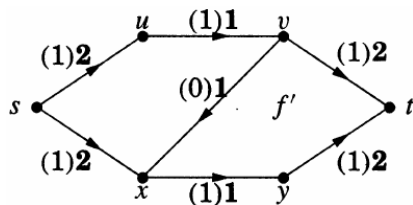
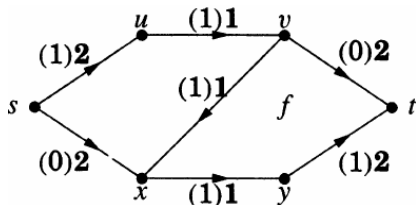
When f is a feasible flow, an **f -augmenting path** is a $s \sim t$ path P in the underlying graph such that for each edge $e \in E(P)$,

- (a) if P follows e in the forward direction, then $f(e) < c(e)$;
- (b) if P follows e in the backward direction, then $f(e) > 0$.

Definition (f -augmenting Paths)

Let P be an f -augmenting path.

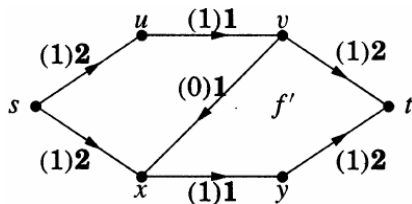
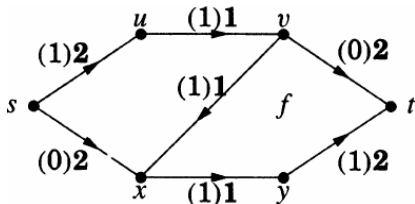
$$\epsilon(e) = \begin{cases} c(e) - f(e) & \text{if } e \text{ is forward on } P \\ f(e) & \text{if } e \text{ is backward on } P \end{cases}$$



Definition (f -augmenting Paths)

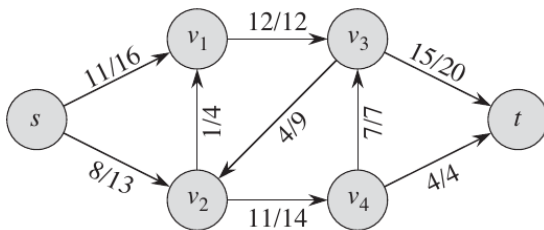
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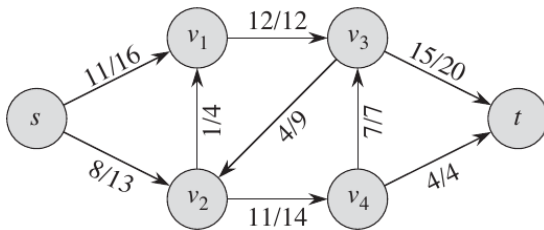
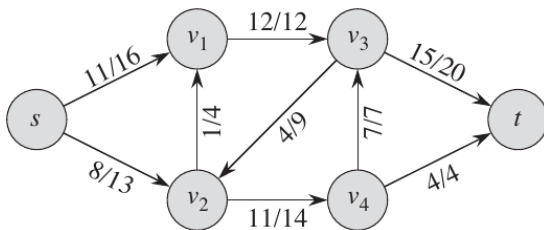
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An f -augmenting path leads to a flow with **larger** value.

$$\min_{e \in E(P)} \epsilon(e)$$

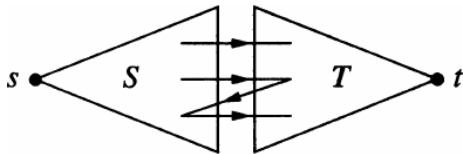


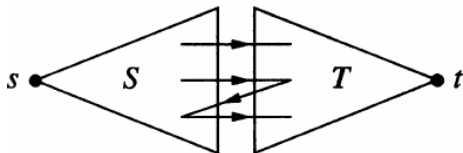


Definition (Source/Sink Cut (割))

In a network, a **source/sink cut** $[S, T]$ consists of the edges **from** a **source set** S **to** a **sink set** T , where

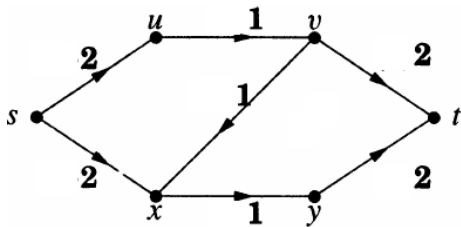
$$(T = V - S) \wedge (s \in S) \wedge (t \in T)$$

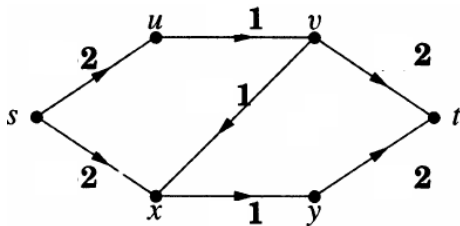




Definition (Capacity of Cut (割的容量))

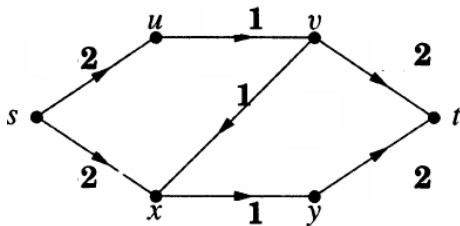
$$\text{cap}(S, T) = \sum_{u \in S, v \in T, uv \in E} c(u, v)$$





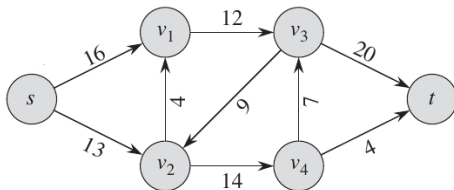
Definition (Minimum Cut (最小割))

A **minimum cut** is a **cut** of minimum value.



Definition (Minimum Cut (最小割))

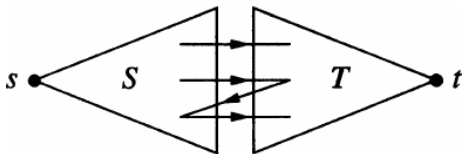
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Theorem (Weak Duality (弱对偶定理))

Let f be any feasible *flow* and $[S, T]$ be any source/sink *cut*.

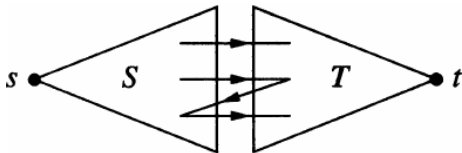
$$\text{val}(f) \leq \text{cap}(S, T).$$



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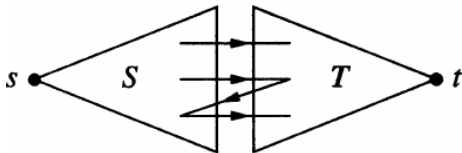


$$\text{val}(f) = f^+(S) - f^-(S)$$

Theorem (Weak Duality (弱对偶定理))

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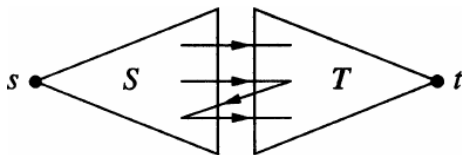


$$\text{val}(f) = f^+(S) - f^-(S) \leq f^+(S)$$

Theorem (Weak Duality (弱对偶定理))

Let f be any feasible *flow* and $[S, T]$ be any source/sink *cut*.

$$\text{val}(f) \leq \text{cap}(S, T).$$



$$\text{val}(f) = f^+(S) - f^-(S) \leq f^+(S) \leq \text{cap}(S, T)$$

Lemma

$$\max_f \text{val}(f) \leq \min_{[S,T]} \text{cap}(S, T)$$

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What if $\text{val}(f) = \text{cap}(S, T)$ for some flow f and some cut $[S, T]$?

Lemma

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What if $\text{val}(f) = \text{cap}(S, T)$ for some flow f and some cut $[S, T]$?

f is maximum and $[S, T]$ is minimum

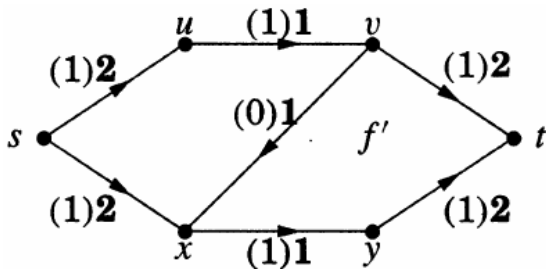
$$\text{val}(f) = f^+(S) - f^-(S) = f^+(S) = \text{cap}(S, T)$$

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$$f^{-1}(S) = 0 \wedge f^+(S) = \text{cap}(S, T)$$

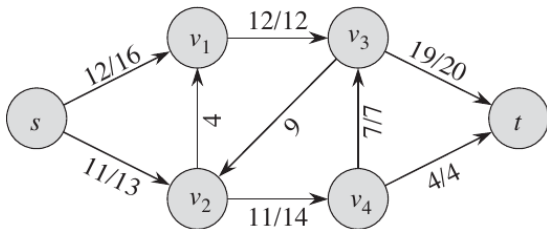
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Theorem (Max-flow Min-cut Theorem (Ford and Fulkerson; 1956))

$$\max_f \text{val}(f) = \min_{[S,T]} \text{cap}(S,T)$$

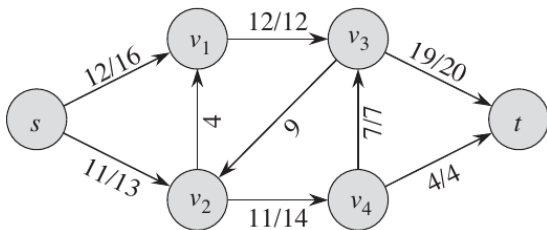
(Strong Duality)

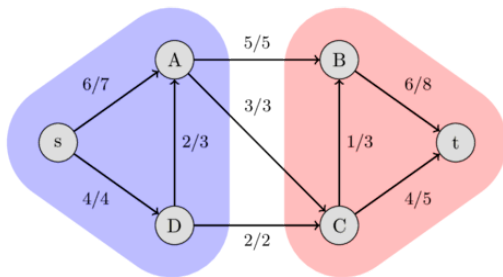
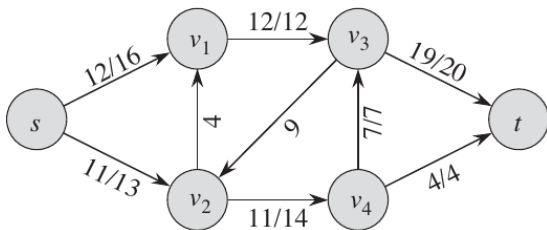


L. R. Ford Jr. (1927 ~ 2017)



D. R. Fulkerson (1924 ~ 1976)





Theorem

A feasible flow f is maximum iff there are no f -augmenting paths.

Theorem

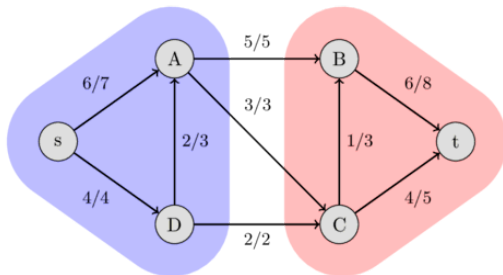
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We construct a cut $[S, T]$ with $\text{val}(f) = \text{cap}(S, T)$.

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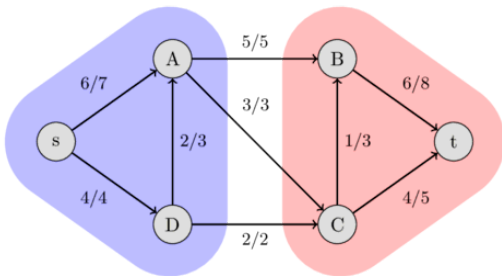
We construct a cut $[S, T]$ with $\text{val}(f) = \text{cap}(S, T)$.



$S \triangleq \{\text{the vertices reachable from } s \text{ along } \text{partial } f\text{-augmenting paths}\}$

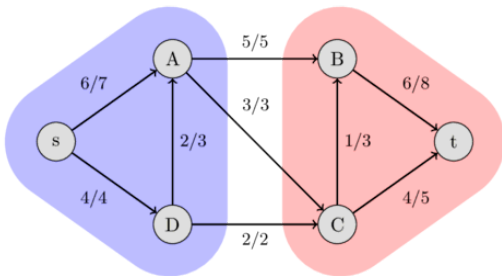
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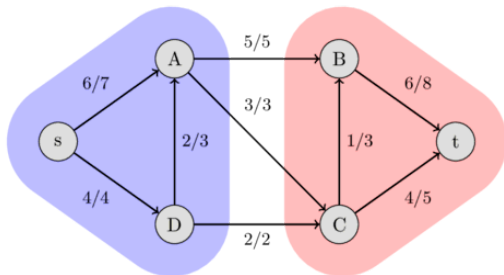
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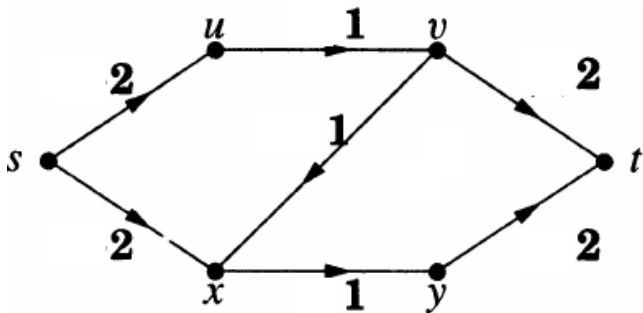
$$\text{val}(f) = f^{+}(S) - f^{-}(S) = f^{+}(S) = \text{cap}(S, T)$$

The Ford-Fulkerson Method

Repeatedly finding f -augmenting paths until no more ones exist.

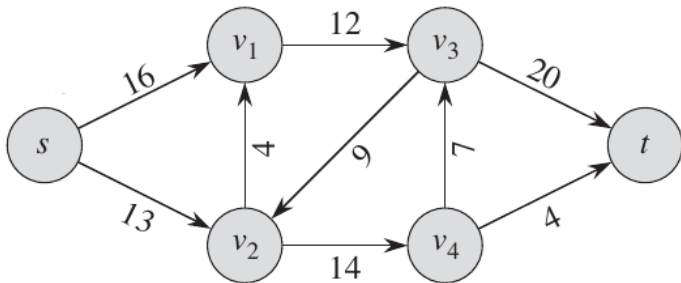
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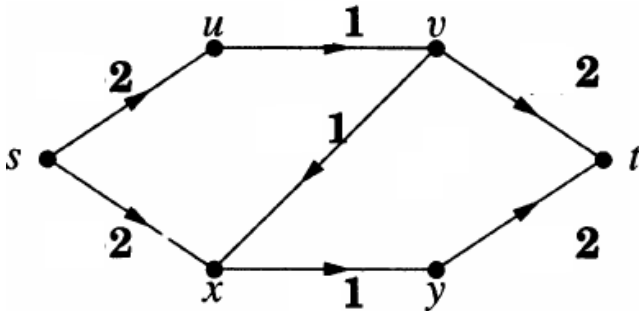


The Edmonds-Karp Algorithm

Using **BFS** (Breadth-first Search) to find f -augmenting paths.

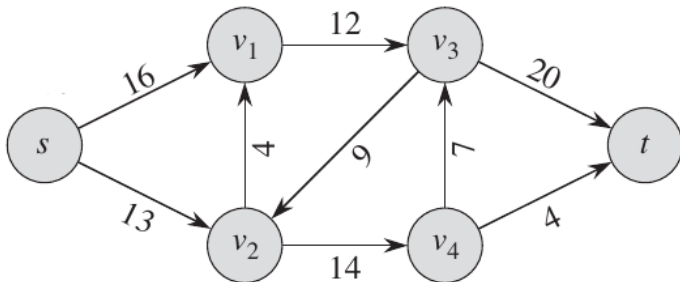
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Theorem (Hall Theorem; 1935)

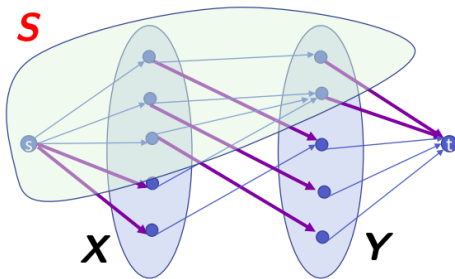
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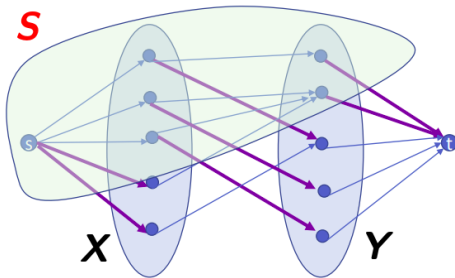


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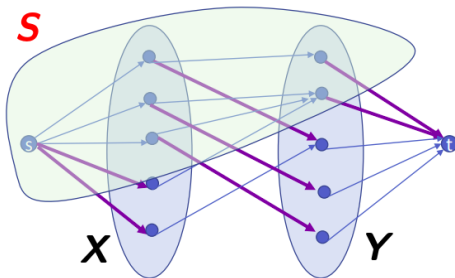
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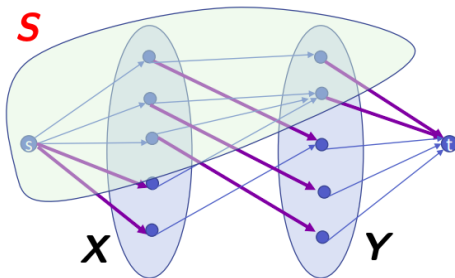
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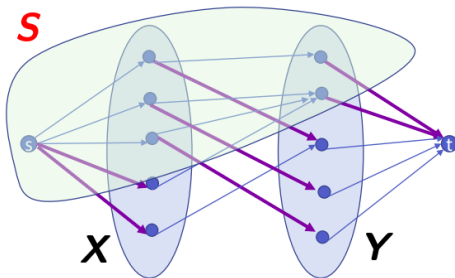


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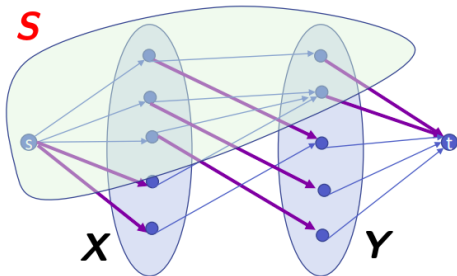
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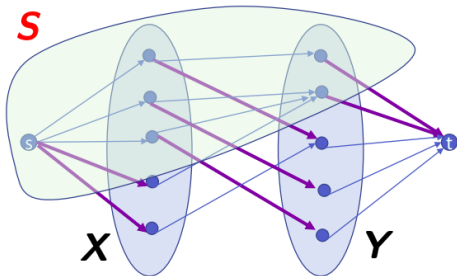
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Let $[S, \bar{S}]$ be a cut. We need to show that $\text{cap}(S, \bar{S}) \geq |X|$.



$$N(S \cap X) \not\subseteq (S \cap Y)$$

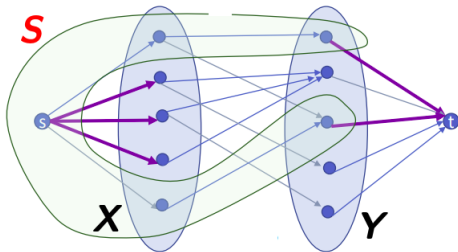
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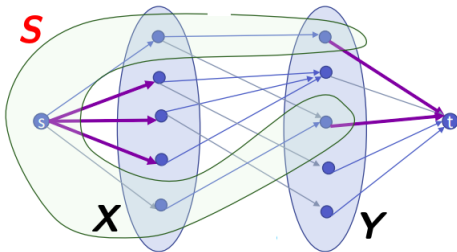
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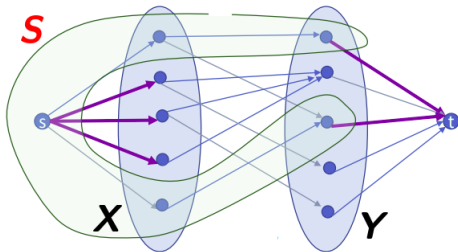


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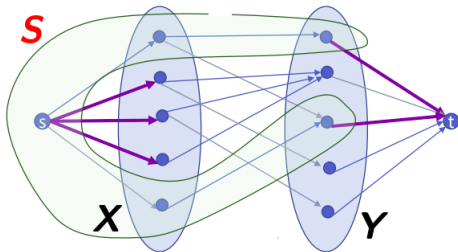
$$\begin{aligned} \text{cap}(S, \bar{S}) &= \sum_{u \in S, v \in \bar{S}} c(u, v) \\ &= \sum_{v \in \bar{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t) \end{aligned}$$

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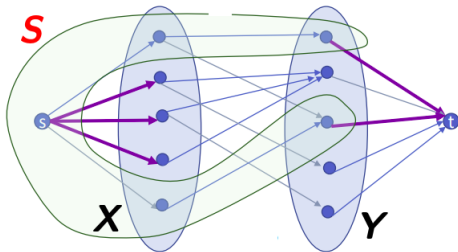
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Theorem (König (1931), Egerváry (1931))

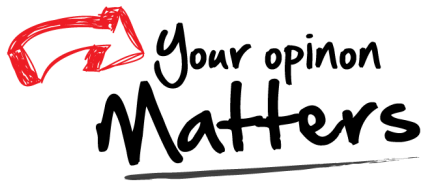
If G is a bipartite graph, then the maximum size of a matching in G equals the minimum size of a vertex cover of G

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Thank
You!



hfwei@hnu.edu.cn